THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET D

Chapters 3 & 4: **A Story of Numbers, Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of 35 in base 8, in the decimal system, is: 1

(a) 28 **(b)** 31 **(c)** 29 **(d)** 30

2. The word 'algorithm' is named after: 1

(a) Fibonacci **(b)** Euclid **(c)** Al-Khwarizmi **(d)** Aryabhata

3. Which Roman symbol stands for 500? 1

(a) D **(b)** C **(c)** M **(d)** L

4. The digits used in base 5 are: 1

(a) 0, 1 **(b)** 1 to 5 **(c)** 0, 1, 2, 3, 4 **(d)** 0 to 5

5. The Indian numerals were transmitted to the Arab world by around: 1

(a) 500 BCE **(b)** 800 CE **(c)** 1500 CE **(d)** 1900 CE

6. Which quadrilateral has two pairs of equal adjacent sides? 1

(a) parallelogram **(b)** kite **(c)** rectangle **(d)** trapezium

7. A parallelogram with equal diagonals is a: 1

(a) rectangle **(b)** trapezium **(c)** rhombus **(d)** kite

8. In a parallelogram ABCD, angle A is 110°. Angle C is: 1

(a) 70° **(b)** 90° **(c)** 250° **(d)** 110°

9. Which of the following cannot have a pair of parallel sides? 1

(a) rectangle **(b)** kite (in general)
(c) trapezium **(d)** rhombus

10. A quadrilateral in which both pairs of opposite angles are equal is a: 1

(a) kite **(b)** scalene quad **(c)** trapezium **(d)** parallelogram

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why did Europeans call Hindu numerals 'Arabic numerals'? 2

12. What is the role of zero in a place value system? 2

13. Find each angle of a rhombus if one angle is 4 times its adjacent angle. 2

14. State why a square is a special rhombus. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Write the numbers 1 to 10 in Roman numerals. 3

16. Explain with a figure idea why a trapezium is not a parallelogram. 3

17. Explain why base 2 needs only two digits. 3

OR

A herd of 37 cows is to be counted in groups of 6. Express the count in base 6. 3

18. Name the quadrilateral formed by joining the midpoints of consecutive sides of a rectangle and state a reason. 3

OR

A square has diagonal 14 cm. Find its area using $\frac{1}{2}d^2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Trace the history of Hindu numerals from India to Europe. 5
(ii) Name two persons involved.

OR

(i) Write the first 10 numbers in base 3. 5
(ii) Which digit is missing?

20. **Case study:** A farmer's field is a trapezium with parallel sides 90 m and 60 m, and height 40 m. 5
(i) Find the area.
(ii) Cost of levelling at ₹5 per m².
(iii) Why is it a trapezium?

OR

(i) What is an interior angle? Find each exterior angle of a rectangle. 5
(ii) Show that the sum of exterior angles of a quadrilateral is 360°.

END OF QUESTION PAPER — Total: 40 Marks

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